FLUXDOCTOR VERSION 1.4

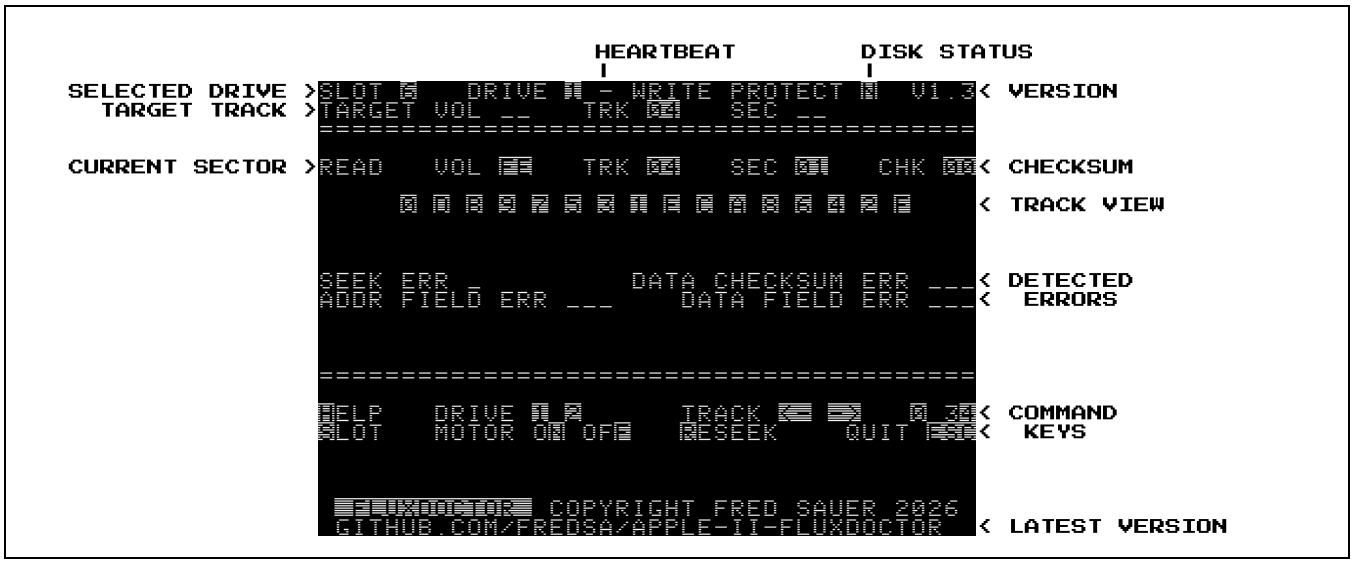
Diagnostic utility for real-time troubleshooting, calibration, and repair of Disk II floppy disk drives.

HARDWARE COMPATIBILITY

* Apple II computer, with at least 20KB RAM installed
* Disk II floppy disk drive

** FLUXDOCTOR MANUAL **

QUICK REFERENCE



COMMAND KEYS

H	Help
S	Select slot 1-6
1, 2	Select drive 1-2
←	Seek to lower track
→	Seek to higher track
0	Seek to track 0
4	Seek to track 34
R	Reseek
N, F	Motor on / off
ESC	Quit

ERROR FLAGS

S - Seek error / unexpected track
M - Missing address or data field
K - Invalid checksum
E - Invalid epilogue, bad byte(s)

FEATURES

* Fast, written in 6502 assembly.
* Runs native on Apple II computers.
* Supports 20KB, 24KB, 32KB, 36KB and 48KB systems.
* Isolates intermittent electromechanical faults.
* Identifies low quality disks.
* Key commands provide operator direct control of stepper and spindle motors.
* Continuous track sampling eases tedious drive diagnosis and repair work.
* Dynamic display indicates sectors read and soft errors normally masked by DOS.
* Autostart allows bench testing without an operational keyboard.
* Cassette loading permits servicing when diskette bootstrap is inoperative.
* Format and write disks out of thin air, thanks to insta-disk.
* Insta-disk formats and writes disks out of thin air.

SECTOR DISPLAY AND STATUS

* In a properly aligned drive, all 16 sectors of a track are read during each rotation. At 300 RPM (200 ms/rev), the sector map updates five times a second.
* DOS 3.3 interleaved diskettes produce a progressive sweep from left to right.
* Read errors will disrupt the sequence. Press 'R' to reinitialize the sector map and clear all error flags.
* Pascal/ProDOS interleaved media will not produce a sweep pattern due to logical sector skew differences.
* Write protect sensor is queried on each seek step. Press 'R' to resample status.

DIAGNOSTIC TOOLS

The experienced repair technician uses factory recorded alignment diskettes. The technician may also employ diskettes formatted on a known-good 48 TPI drive.

Diskettes recorded on 96 TPI drives will induce a low amplitude signal on 48 TPI heads. Visible soft errors will result.

Overwriting a 48 TPI track on a 96 TPI drive leaves unerased margins that will further exacerbate the problem.

For increased reliability, a neodymium magnet may be passed over a diskette in its jacket to thoroughly erase residual magnetic flux prior to reformatting.

EMI CONSIDERATIONS

Electromagnetic interference (EMI) from the horizontal flyback transformer of a CRT display or television receiver will induce noise into Disk II read heads.

Interference is most pronounced adjacent to the transformer, commonly located at the rear corner of the monitor chassis. Poor read performance and visible soft errors will be seen on working systems.

Narrow-track (96 TPI) recorded media produces very low amplitude signals when read by wide-track (48 TPI) heads. Unacceptable error rates will be seen in the presence of flyback interference.

Maintain at least 12 inches separation between Disk II and the CRT flyback.

PROGRAM LOADING OPTIONS

DISK - DOS 3.3 floppy disk image (140kB)

> Autostart disk images are provided in *.DO format. A Greazeweazle may be used to create a copy of FLUXDOCTOR from a modern computer.

TAPE - cassette tape audio file

> Auto-run audio tape files are provided in *.WAV format. A phone or modern computer may be used to PLAY the audio for direct LOAD into your Apple II computer via a suitable audio cable.

TYPE-IN - machine language monitor listing

> Users who enjoy doing so may type-in FLUXDOCTOR using the built-in machine language monitor.

INSTA-DISK - self-writing disk audio file

> Format and write FLUXDOCTOR disks just by loading the insta-disk from the provided audio file.

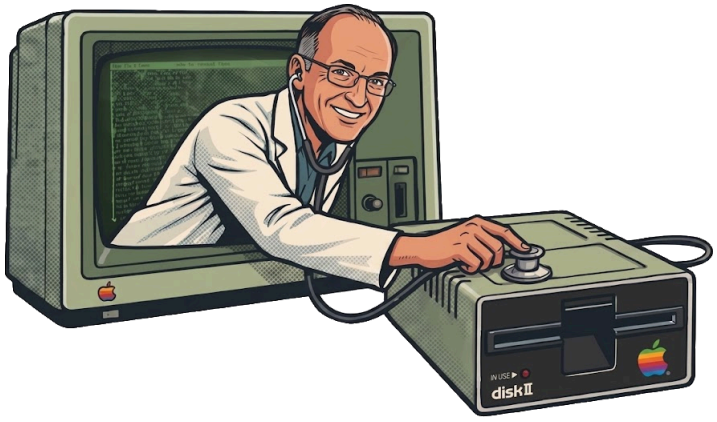
SOFTWARE UPDATES

Autostart diskette images and audio tape diagnostic and repair software may be downloaded at no cost to the user from:

https://github.com/fredsa/apple-ii-fluxdoctor

Don't delay, download the latest version of fluxdoctor today!

CUSTOMER CORRESPONDENCE



The author of this program requests your feedback. Inquiries may be directed to fredsa@gmail.com or posted to the public issue tracker at:

https://github.com/fredsa/apple-ii-fluxdoctor

Hearing directly from users like you serves as encouragement and provides motivation for future work.

The author looks forward to reading your comments and watching your live television broadcasts.

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